MADHAV MERUVA

Software Engineer | Backend & AI/ML
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SUMMARY

Computer Science undergraduate specializing in **Backend Engineering and AI/ML systems**, with hands-on experience building **LLM-powered applications, vector search systems, and scalable APIs**. Improved system performance by **30% through caching and optimization techniques**, and developed end-to-end ML pipelines with real-time inference. Seeking Software Engineering / AI-ML internship roles.

EDUCATION

Keshav Memorial Institute of Technology (KMIT)
Bachelor of Technology, Computer Science Engineering | 2023 - 2027 | CGPA - 8.5

Sri Chaitanya Junior College
Intermediate (MPC) | 2021-2023 | GPA – 9.8

PROJECTS

AI Startup Idea Validator (Full-Stack) | Aug 2025 – Oct 2025
Live: startupideavalidator.vercel.app | **GitHub:** github.com/mad674/startupideavalidator

- Built a **scalable LLM-powered** platform for real-time startup evaluation.
- Designed **modular backend architecture** with scalable services and optimized API performance
- Improved API response time by **~30%** using Redis caching, idempotent APIs, and pagination strategies
- Implemented **semantic search** using Vector DB and LangChain, improving relevance for AI-generated insights
- **Tech Stack:** React.js, Node.js (Express), FastAPI, MongoDB, PineconeDB, Redis, LangChain, OpenAI API, JWT

Financial Question Answering System (FinQA) | Feb 2025 - Jun 2025
Live: financegptapp.netlify.app **| GitHub:** github.com/mad674/qa

- Built a **neuro-symbolic QA system** using a BERT retriever and LSTM-based program generator.
- Engineered **end-to-end ML pipeline** including preprocessing, model training, evaluation, and inference
- Achieved **~80% accuracy** with interpretable program-step reasoning and symbolic execution.
- **Tech Stack:** Flutter, FastAPI, PyTorch, Firebase

Jewellery Design Pattern Generator (Pix2Pix GAN) | Aug 2024 - Dec 2024
Live: jewelrydesignpatterngen.netlify.app **| GitHub:** github.com/mad674/mlmodel

- Built a **Pix2Pix GAN-based system for sketch-to-image generation** with real-time inference capabilities
- Developed a **real-time pipeline for generating design previews** from user-uploaded sketches
- Improved output quality through **dataset preprocessing** and training optimization techniques
- **Tech Stack:** React.js, Node.js (Express), JWT, MongoDB, TensorFlow, Flask

SKILLS

- **Languages:** Java, Python, C++, JavaScript
- **Backend:** Node.js (Express), FastAPI, Flask, REST APIs
- **Frontend:** React.js
- **Databases:** MongoDB, MySQL, Vector Databases
- **AI/ML:** PyTorch, TensorFlow, NLP, Transformers, LangChain
- **Tools:** Git, Postman, AWS, Docker, Redis
- **Core Concepts:** Data Structures & Algorithms, OOP, OS

ACHIEVEMENTS AND CERTIFICATES

- Solved **500+ DSA** problems across arrays, graphs, trees, and dynamic programming
- Secured a **top team award** at the KMIT Project Expo
- Participated in **national-level hackathons** (Vasavi & VNRVJIET)